

Day 1 — The Sun’s Yearly Path

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: observe that sunrise direction changes; introduce the ecliptic.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Sūrya* (IAST: Sūrya; lit. “the Sun”)

Primary source (today’s reading)

From *Sūrya-siddhānta* (c. 5th–10th c. CE) — see `sources.md`.

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **The observation.** Project a year-long animation of sunrise direction (or show a static composite of 12 sunrise photos taken from the same spot, one per month). “*The Sun rises in roughly the east — but the EXACT direction shifts north and south through the year.*”
2. **Define the ecliptic.** The Sun’s apparent path against the background stars over one year. A great circle on the celestial sphere, tilted $\sim 23.5^\circ$ from the celestial equator.
3. **Why does it shift?** Earth’s rotational axis is tilted $\sim 23.5^\circ$ relative to its orbital plane. As Earth orbits the Sun, this tilt projects onto our sky as the Sun rising/setting north of due-east in summer (NH), south in winter.
4. **Source.** *Sūrya-siddhānta* ch. 2 — “*sūryasya gatiḥ*” — gives a mean and true solar motion model. Specifies daily and annual displacement on the ecliptic.
5. **Independent task.** Each student looks up their city’s sunrise direction today vs 6 months from today (web). Note the difference in compass bearing.

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces The Sun's Yearly Path.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).